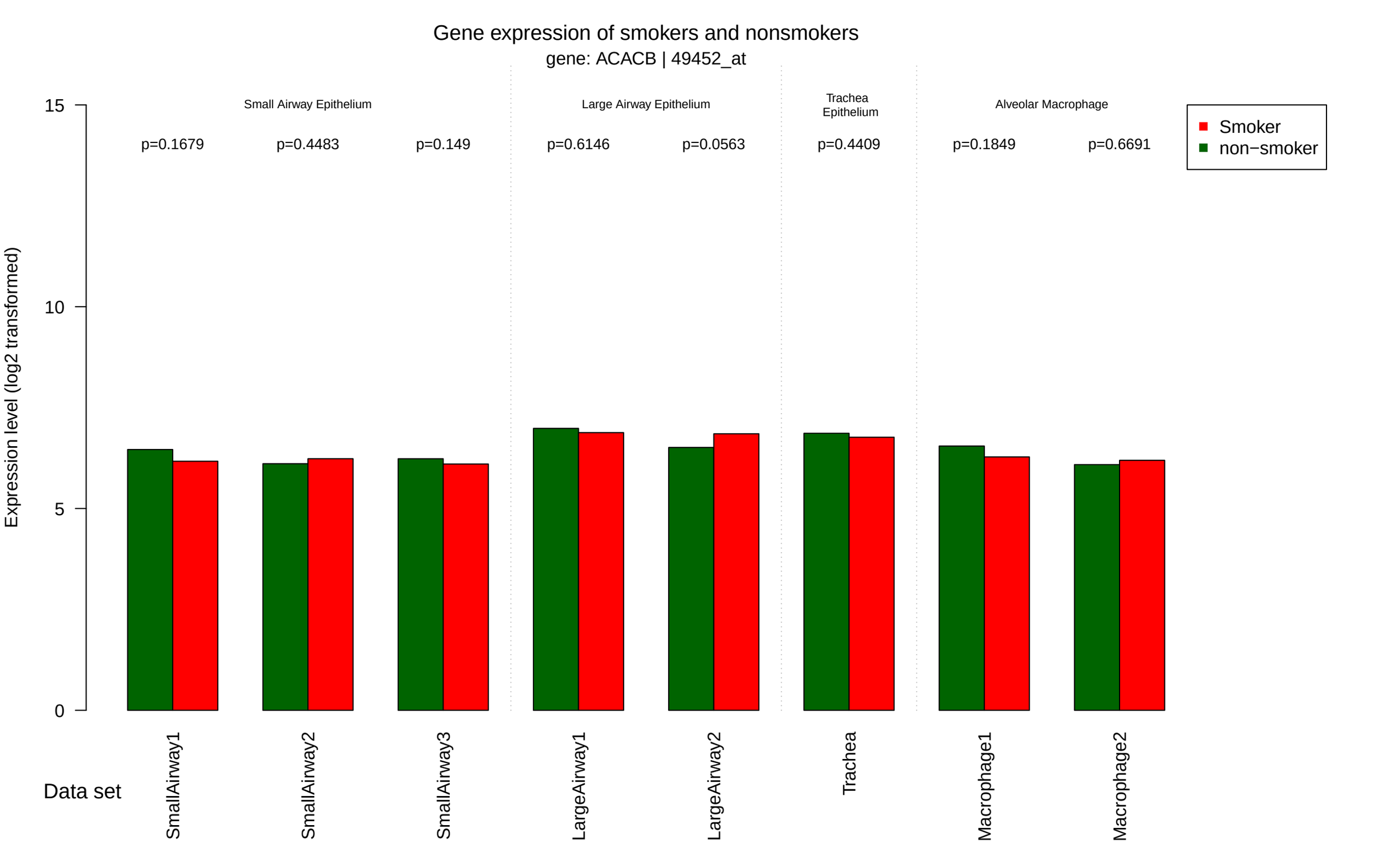
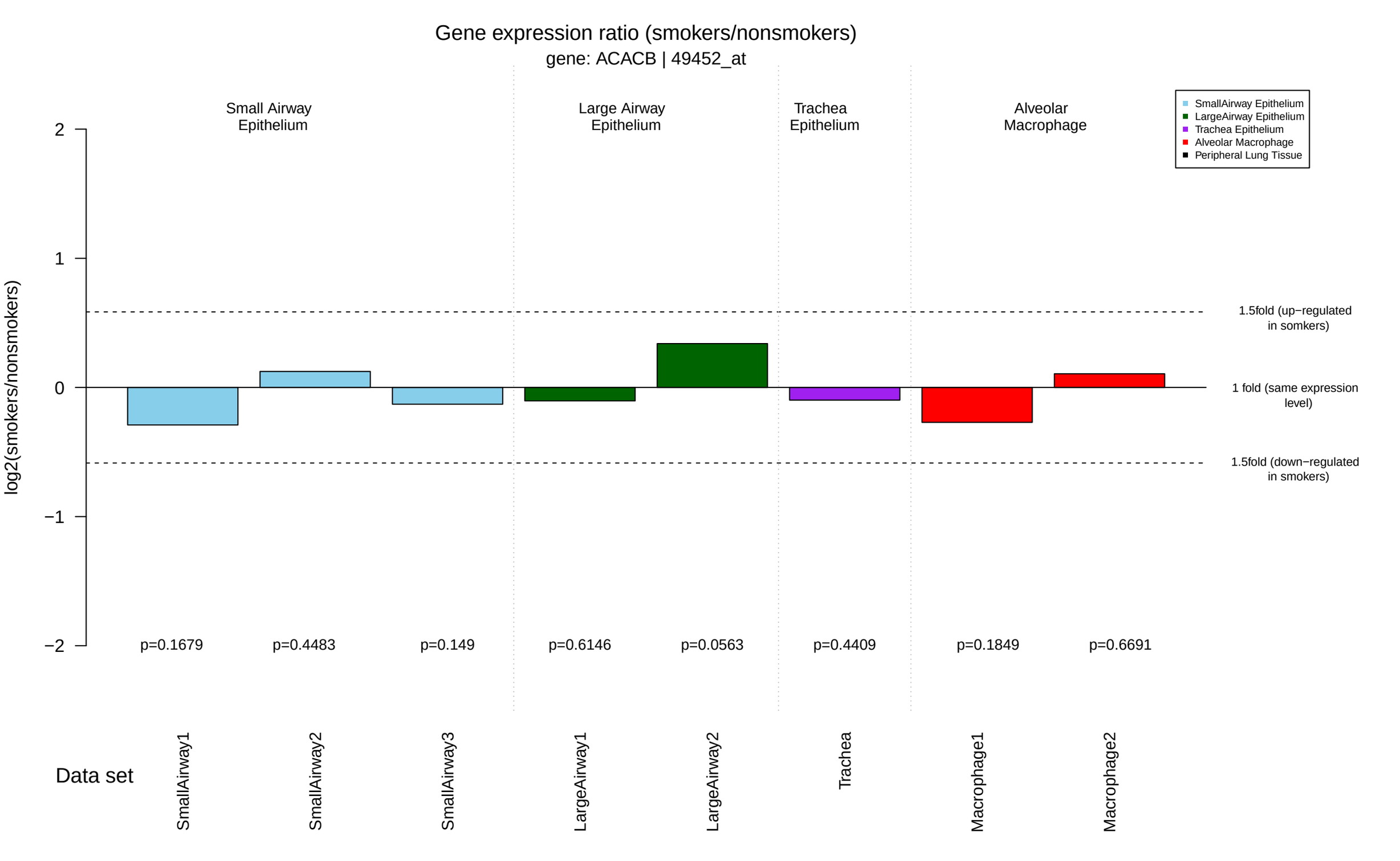
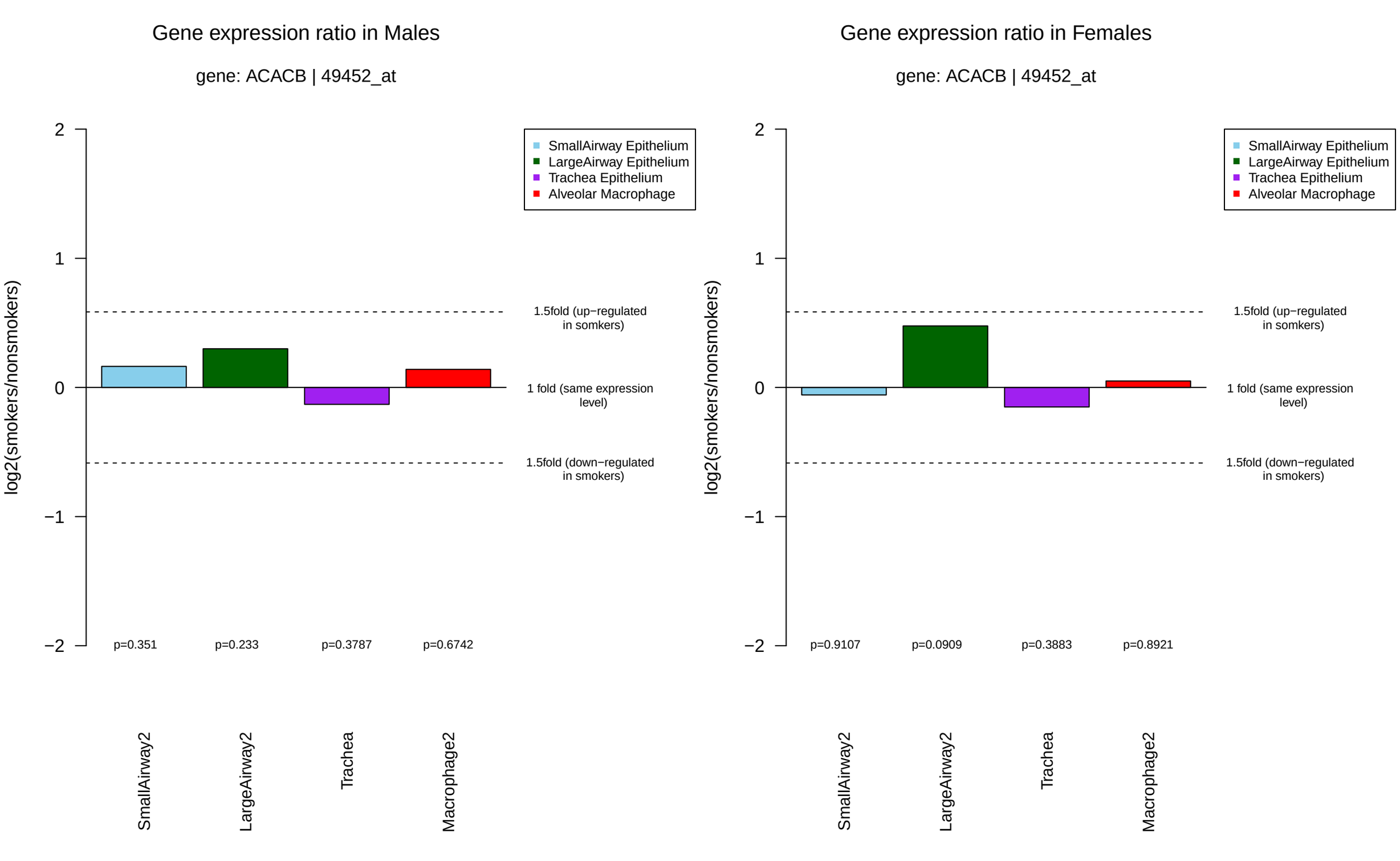
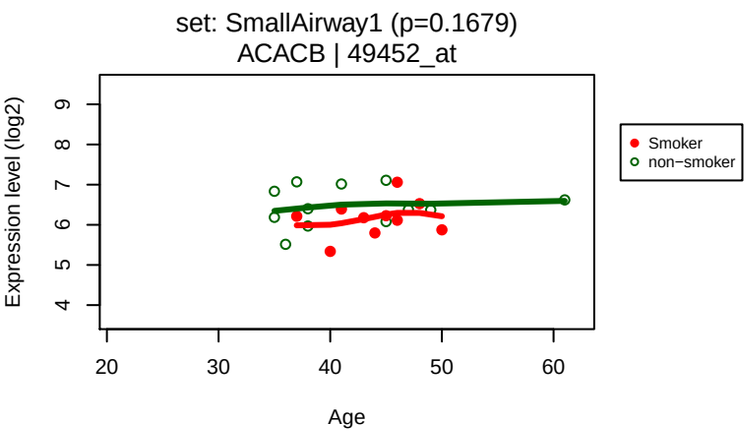
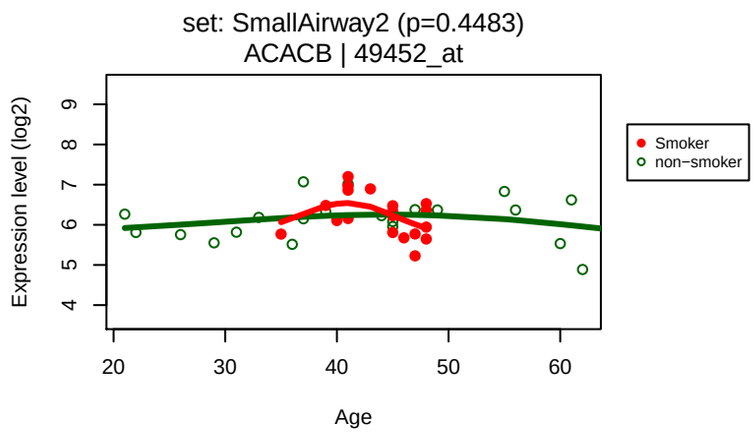
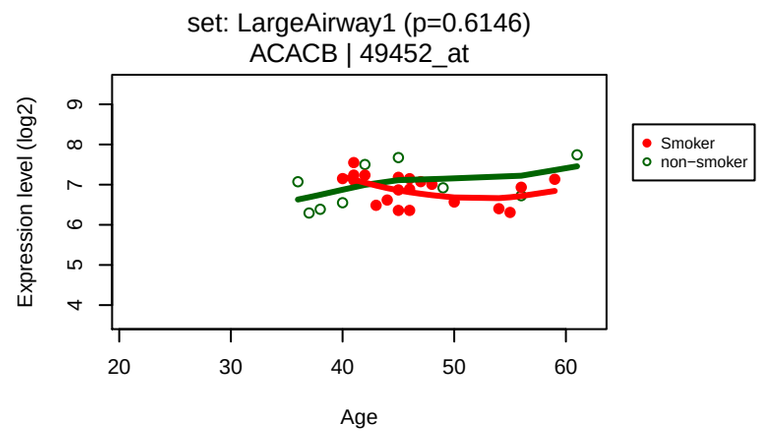

Gene expression of smokers and nonsmokers
gene: ACACB | 49452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1679
p=0.4483
p=0.149
p=0.6146
p=0.0563
p=0.4409
p=0.1849
p=0.6691
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

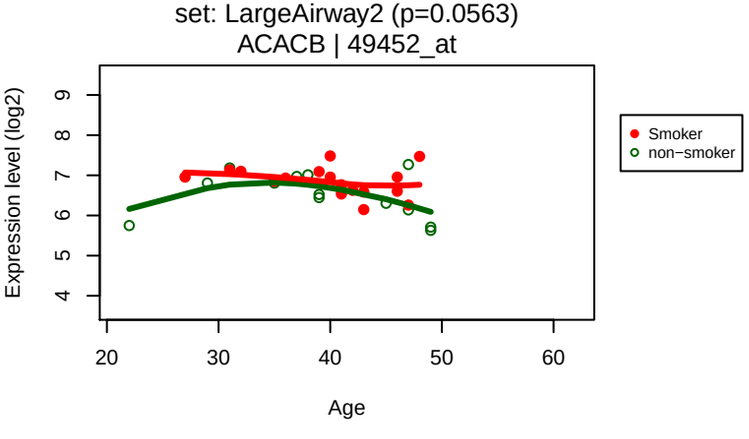

Gene expression ratio (smokers/nonsmokers)
gene: ACACB | 49452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1679
p=0.4483
p=0.149
p=0.6146
p=0.0563
p=0.4409
p=0.1849
p=0.6691
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

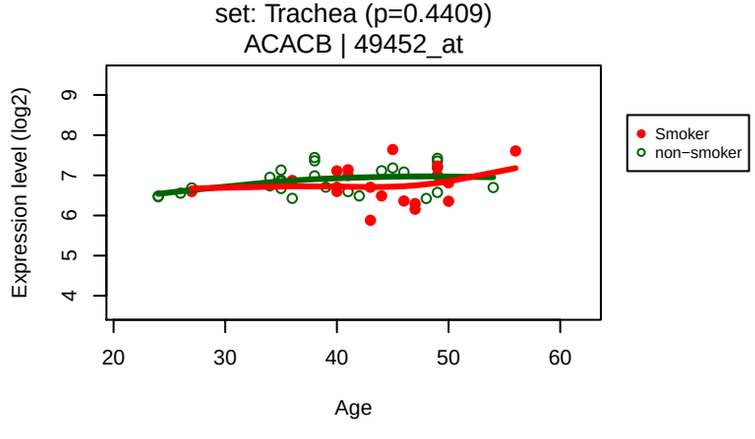

Gene expression ratio in Males
gene: ACACB | 49452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.351
p=0.233
p=0.3787
p=0.6742
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACACB | 49452_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9107
p=0.0909
p=0.3883
p=0.8921
SmallAirway2
LargeAirway2
Trachea
Macrophage2

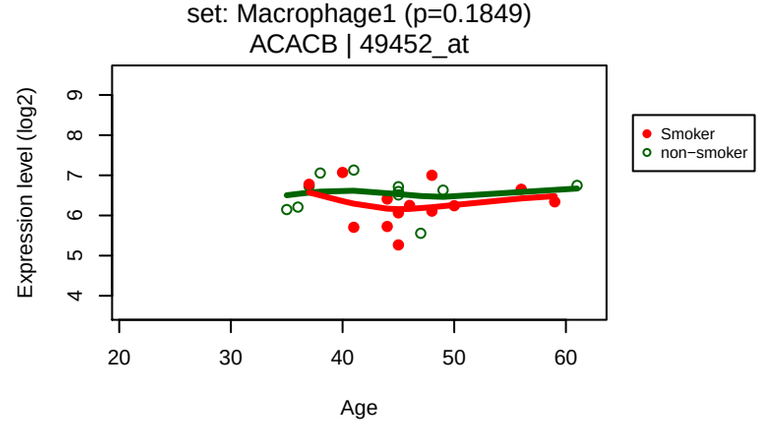

set: SmallAirway1 (p=0.1679)
ACACB | 49452_at
Expression level (log2)
Age
Smoker
non−smoker

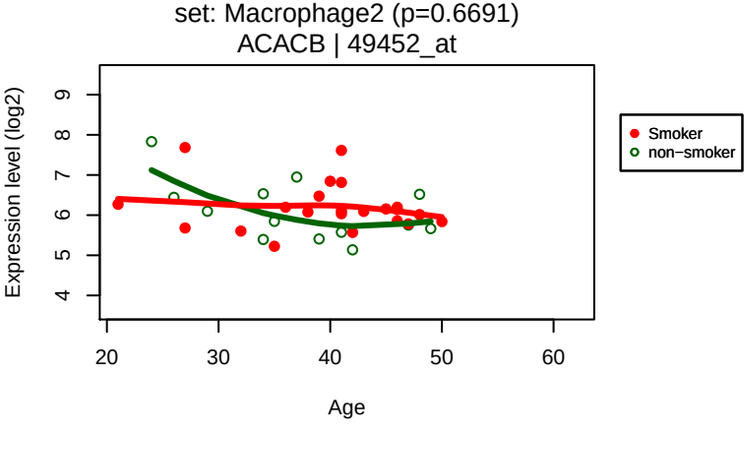

set: SmallAirway2 (p=0.4483)
ACACB | 49452_at
Expression level (log2)
Age
Smoker
non−smoker

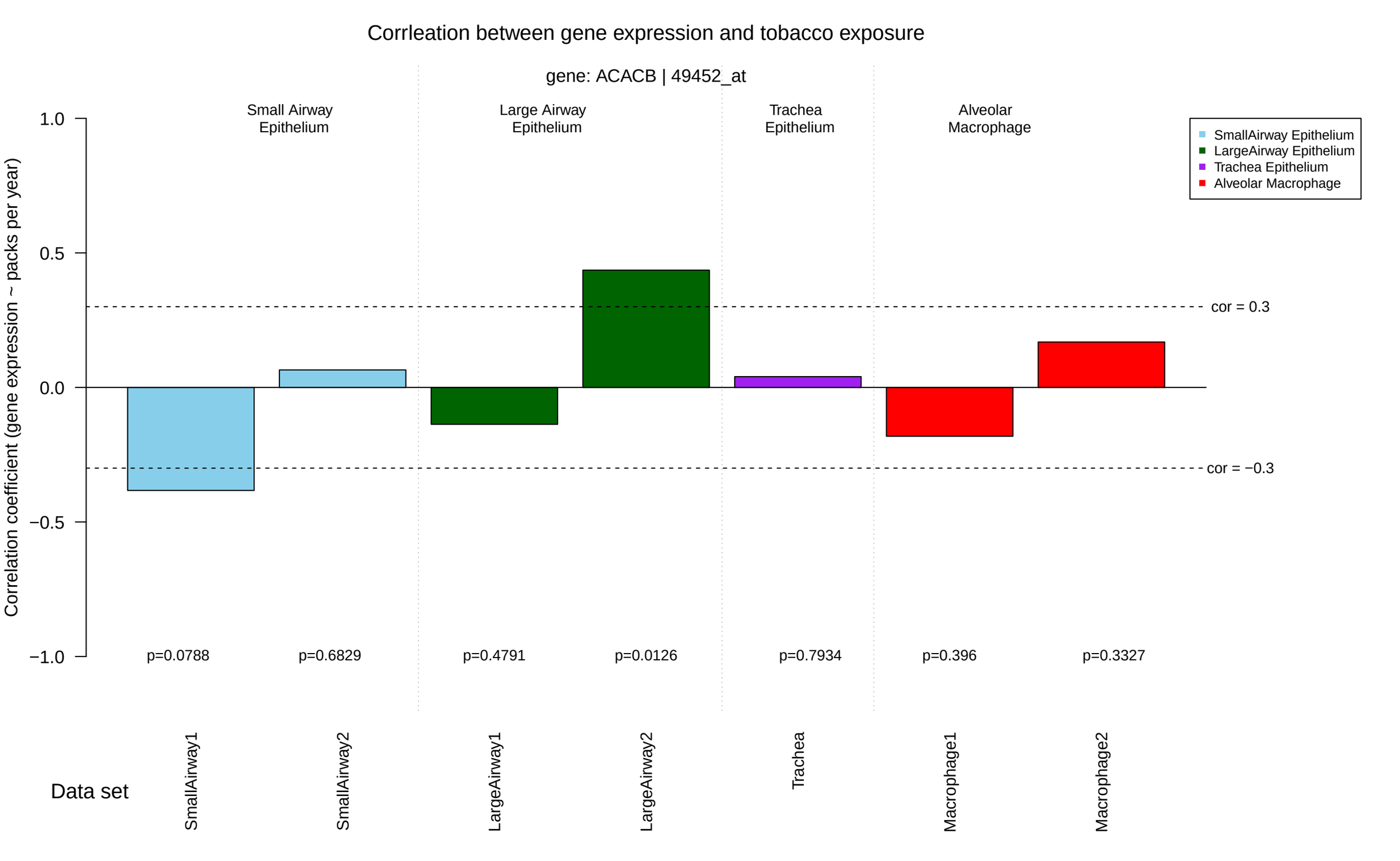

set: LargeAirway1 (p=0.6146)
ACACB | 49452_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0563)
ACACB | 49452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4409)
ACACB | 49452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1849)
ACACB | 49452_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6691)
ACACB | 49452_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACACB | 49452_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0788
p=0.6829
p=0.4791
p=0.0126
p=0.7934
p=0.396
p=0.3327
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2